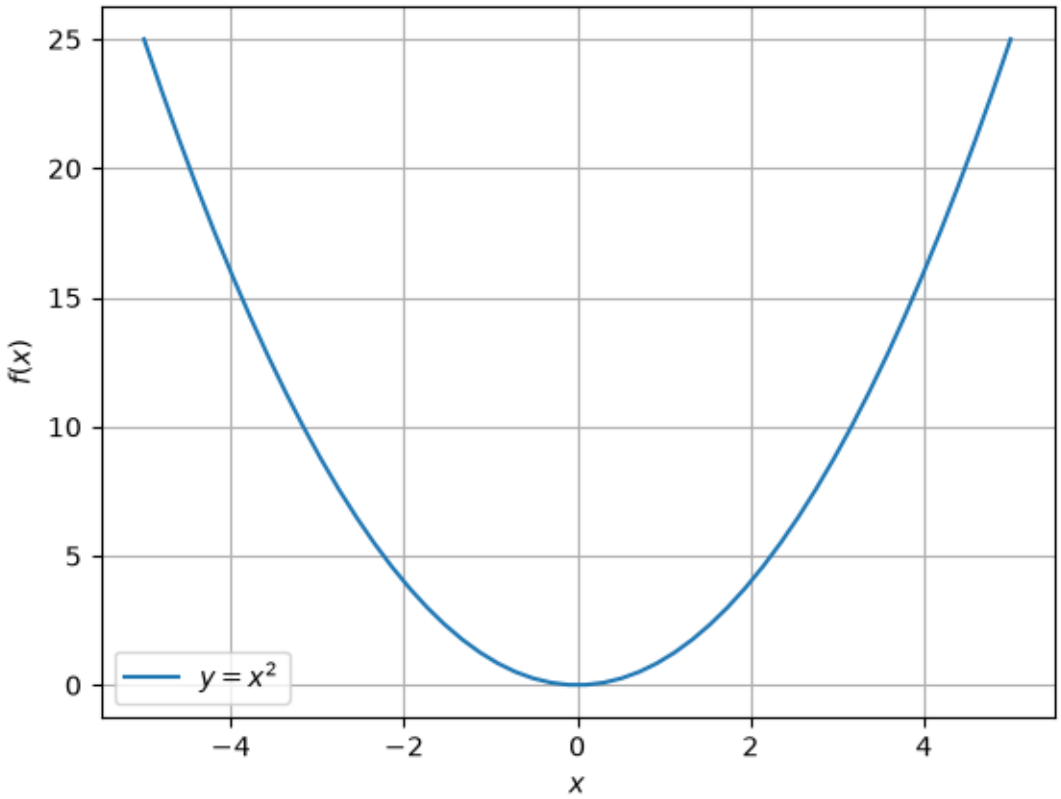
tarefa_matplotlib

October 9, 2026

1. gráfico de uma função

[88]:
```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-5, 5)
plt.plot(x, x**2, label=r'$y=x^2$')
plt.xlabel(r'$x$')
plt.ylabel(r'$f(x)$')
plt.grid()
plt.legend()
plt.show()
```

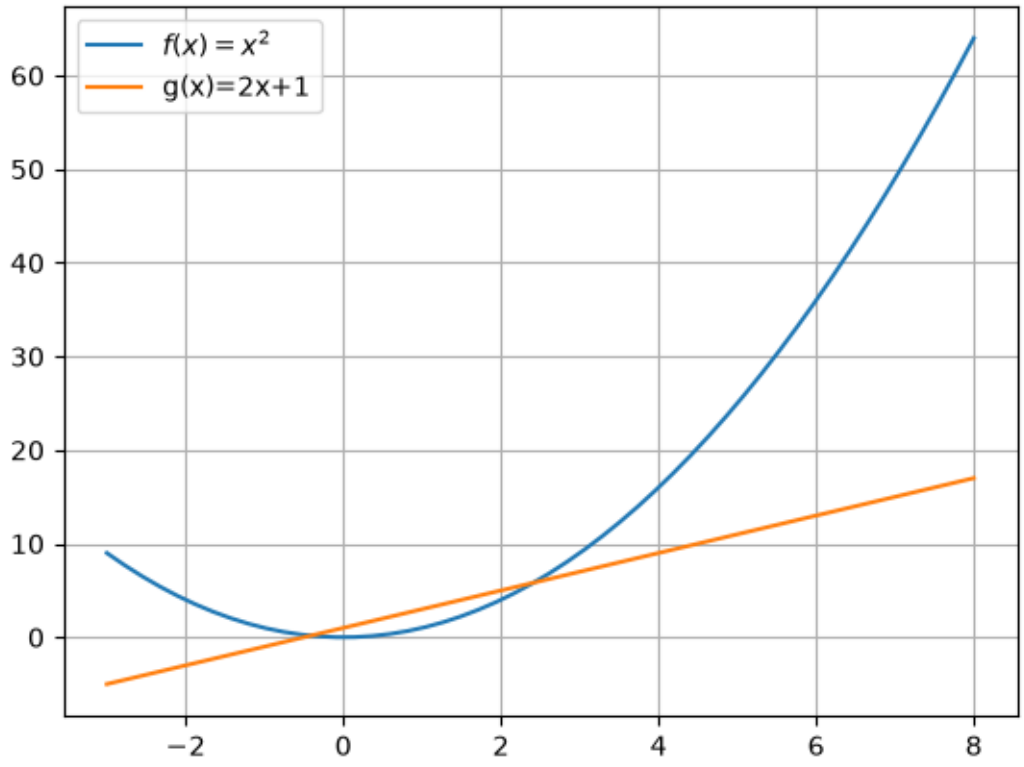


2. duas funções

[5]:
```
x = np.linspace(-3, 8)

plt.plot(x, x**2, label=r'$f(x)=x^2$')
plt.plot(x, 2*x + 1, label=r'g(x)=2x+1')

plt.legend()
plt.grid()
plt.show()
```



3. gráfico de seno e cosseno

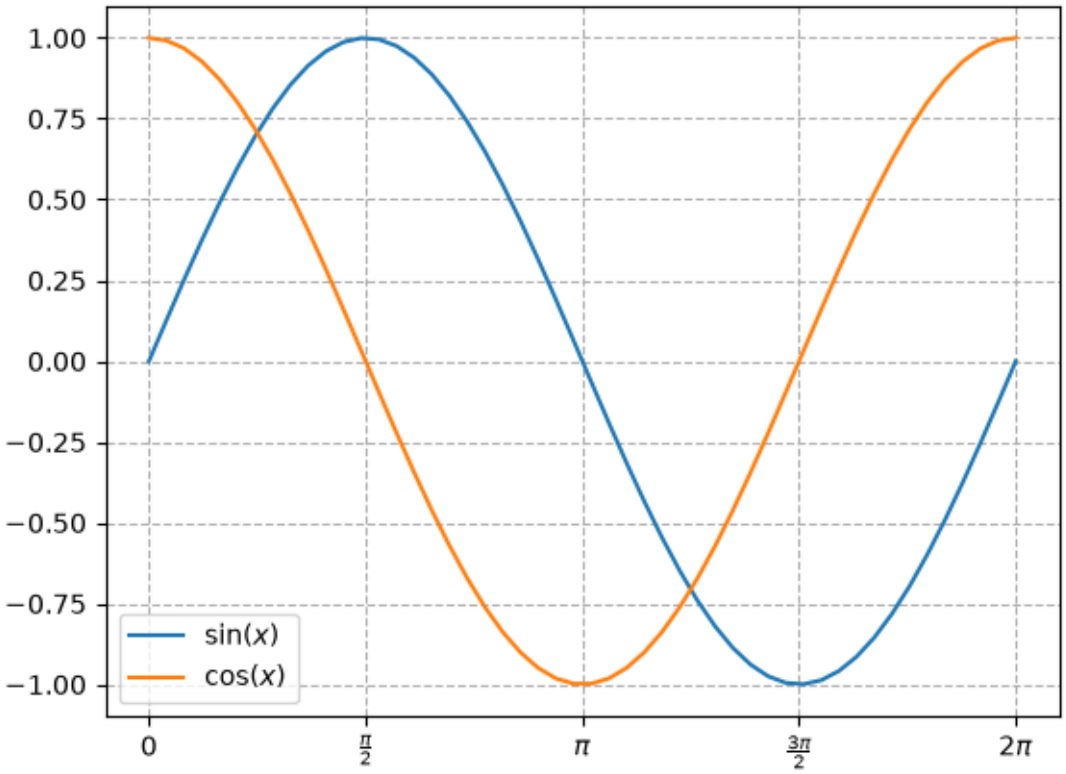
[10]:
```
x = np.linspace(0, 2*np.pi)

plt.plot(x, np.sin(x), label=r'$\sin(x)$')
plt.plot(x, np.cos(x), label=r'$\cos(x)$')

tick_pos = np.arange(0, 2.1 * np.pi, np.pi / 2)

tick_labels = [
    r'0', r'$\frac{\pi}{2}$', r'$\pi$', r'$\frac{3\pi}{2}$', r'$2\pi$'
]
```

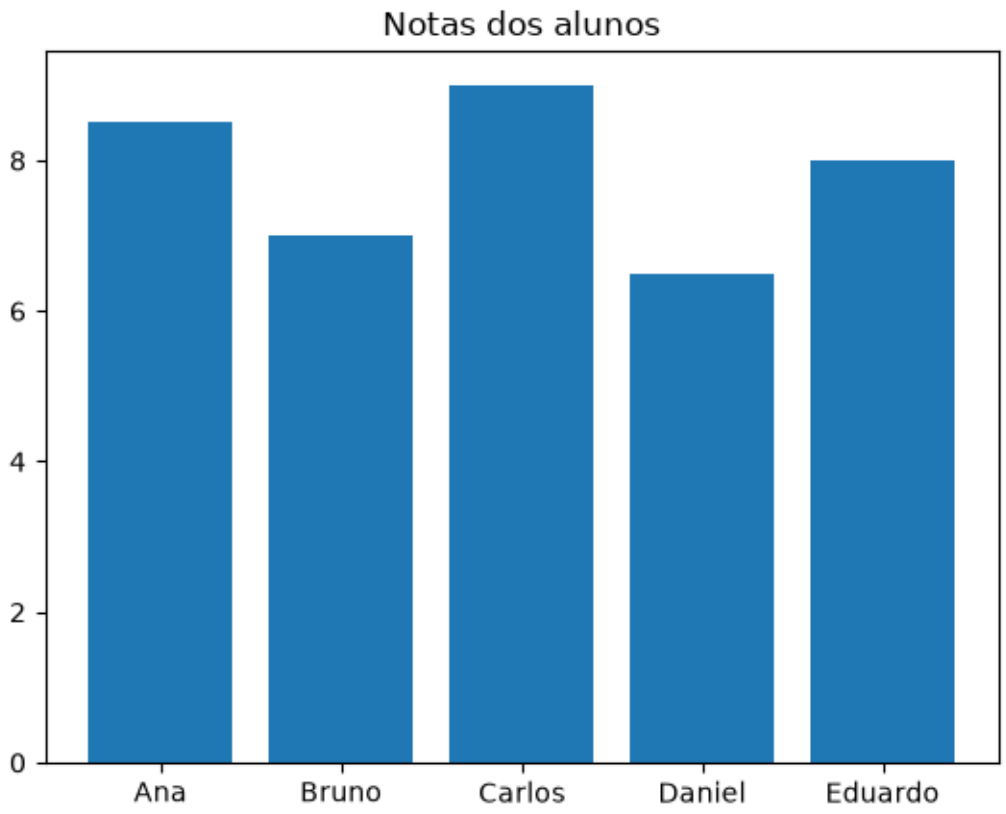
```
plt.xticks(tick_pos, tick_labels)
plt.grid(True, linestyle='--')
plt.legend()
plt.show()
```



4. gráfico de barras

[21]:
```
quadro_notas = {"Ana": 8.5, "Bruno": 7.0, "Carlos": 9.0, "Daniel": 6.5,␣
 ↪"Eduardo": 8.0}
alunos = list(quadro_notas.keys())
notas = list(quadro_notas.values())

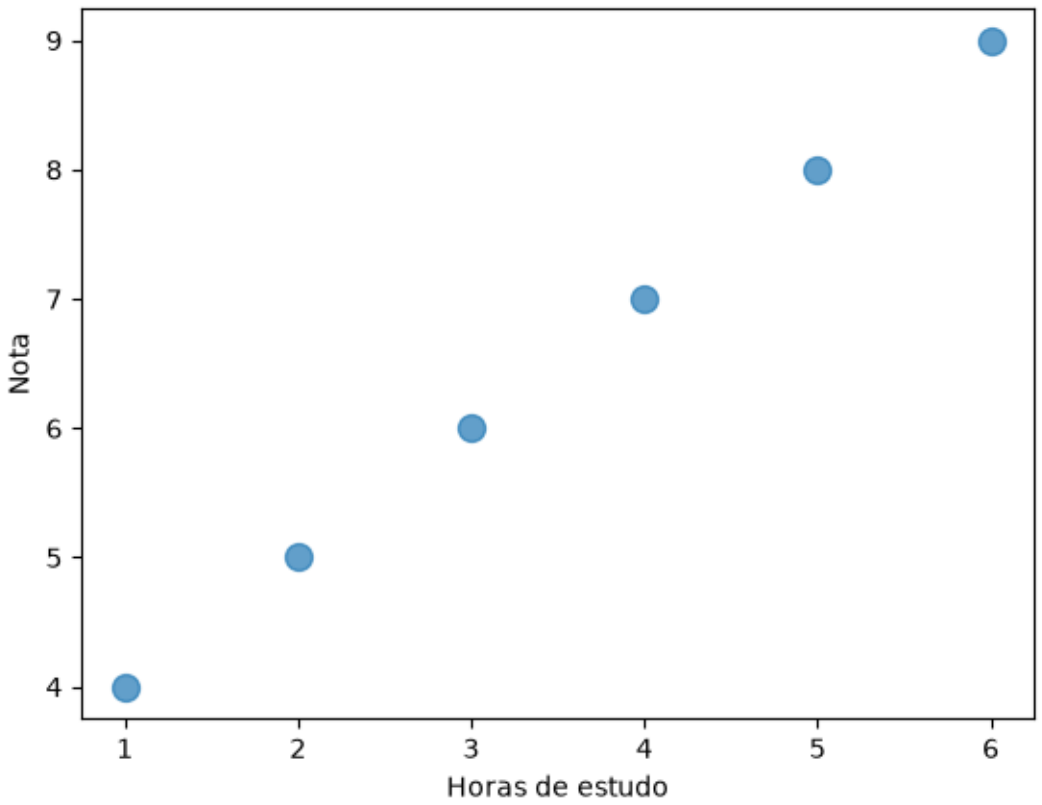
plt.bar(alunos, notas)
plt.title("Notas dos alunos")
plt.show()
```



5. gráfico de dispersão

[24]:
```
horas = [h for h in range(1, 7)]
nota = [n for n in range(4, 10)]

plt.scatter(
    horas, nota,
    s=100,
    marker='o',
    alpha=0.7
)
plt.xlabel("Horas de estudo")
plt.ylabel("Nota")
plt.show()
```



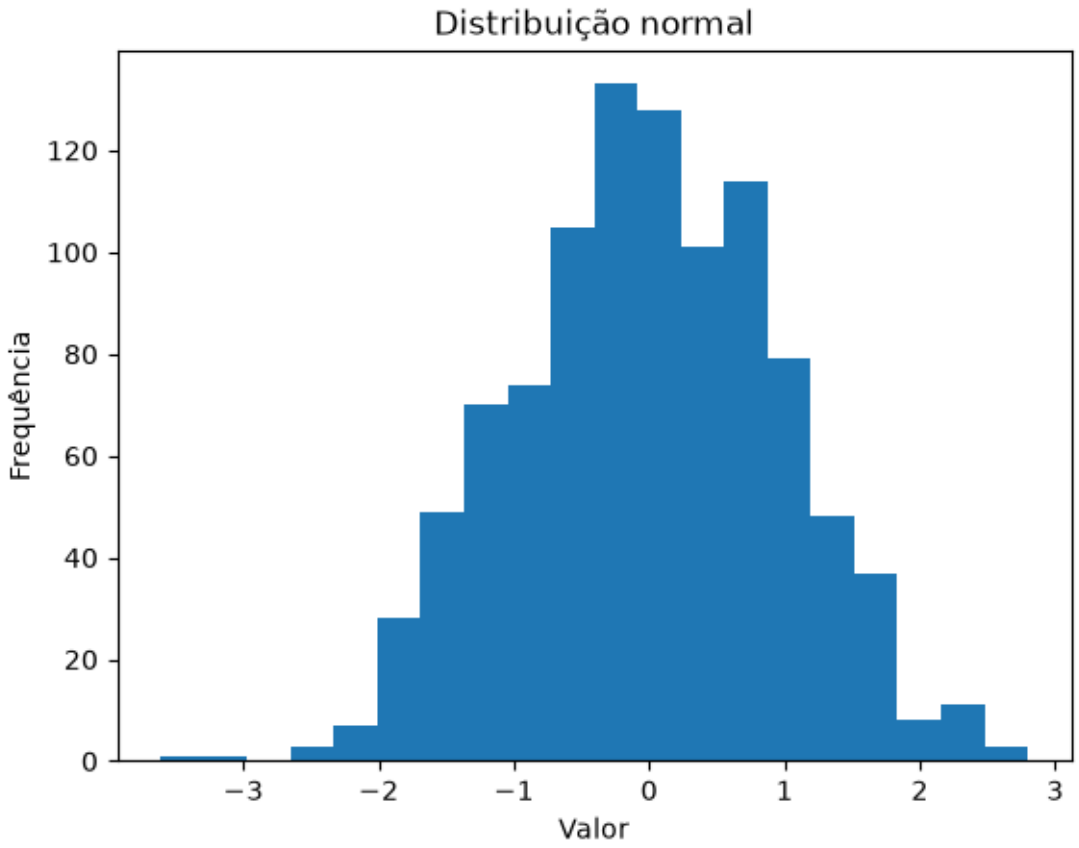
6. histograma

[28]:
```
dados = np.random.normal(0, 1, 1000)

plt.hist(dados, bins=20)

plt.xlabel("Valor")
plt.ylabel("Frequência")
plt.title("Distribuição normal")

plt.show()
```

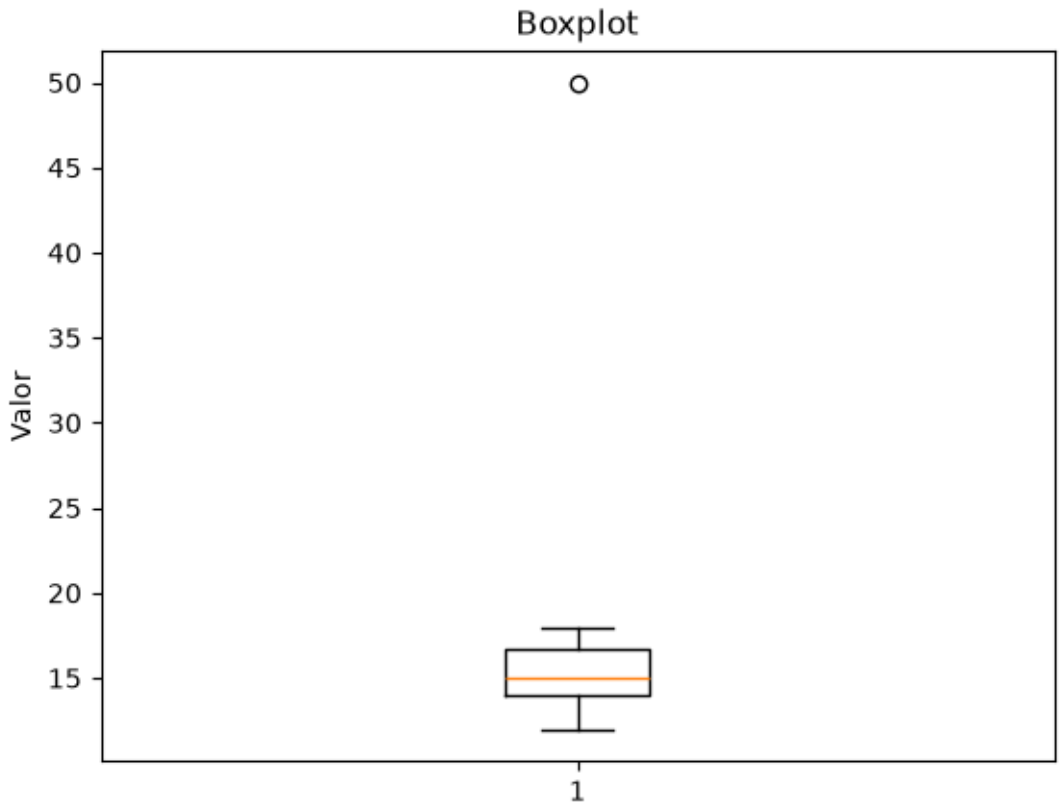


7. boxplot

[29]:
```
tempos = [12, 15, 13, 17, 14, 16, 15, 18, 14, 50]

plt.boxplot(tempos)

plt.title("Boxplot")
plt.ylabel("Valor")
plt.show()
```



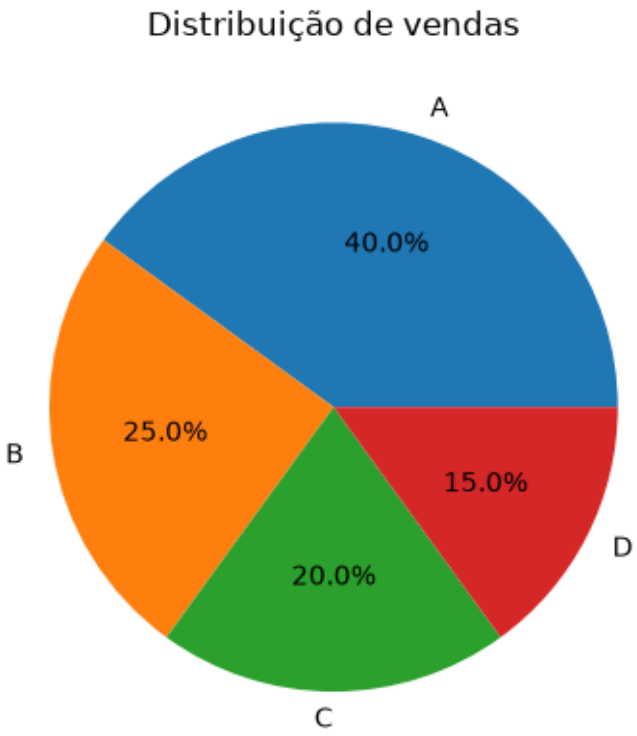
O gráfico `boxplot` nos permite facilmente identificar o *outlier* 50.

8. gráfico de pizza

[36]:
```
vendas = [40, 25, 20, 15]
produtos = ["A", "B", "C","D"]

plt.pie(
    vendas,
    labels=produtos,
    autopct="%1.1f%%"
)

plt.title("Distribuição de vendas")
plt.show()
```

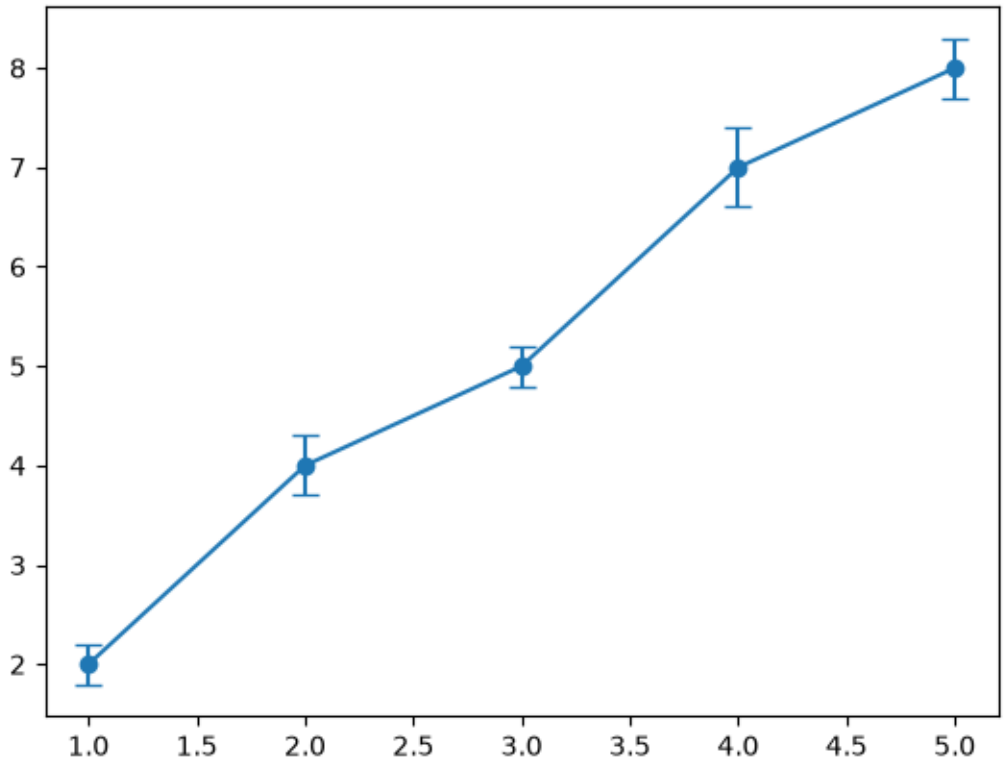


9. gráfico de erro

[37]:
```
x = [1, 2, 3, 4, 5]
y = [2, 4, 5, 7, 8]
erro = [0.2, 0.3, 0.2, 0.4, 0.3]

plt.errorbar(
    x, y,
    yerr=erro,
    fmt="o-",
    capsize=5
)

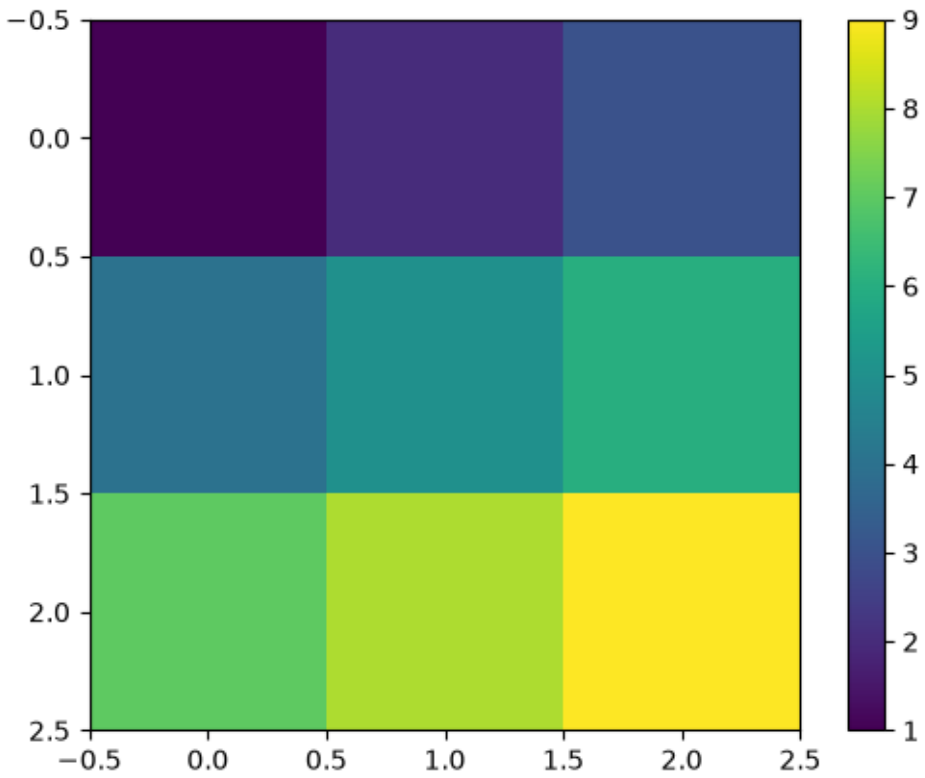
plt.show()
```



10. mapa de calor de uma matriz

[39]:
```
A = np.arange(1, 10).reshape(3, 3)

plt.imshow(A)
plt.colorbar()
plt.show()
```



11. gráfico de contorno

[40]:
```
x = np.linspace(-5, 5)
y = np.linspace(-5, 5)

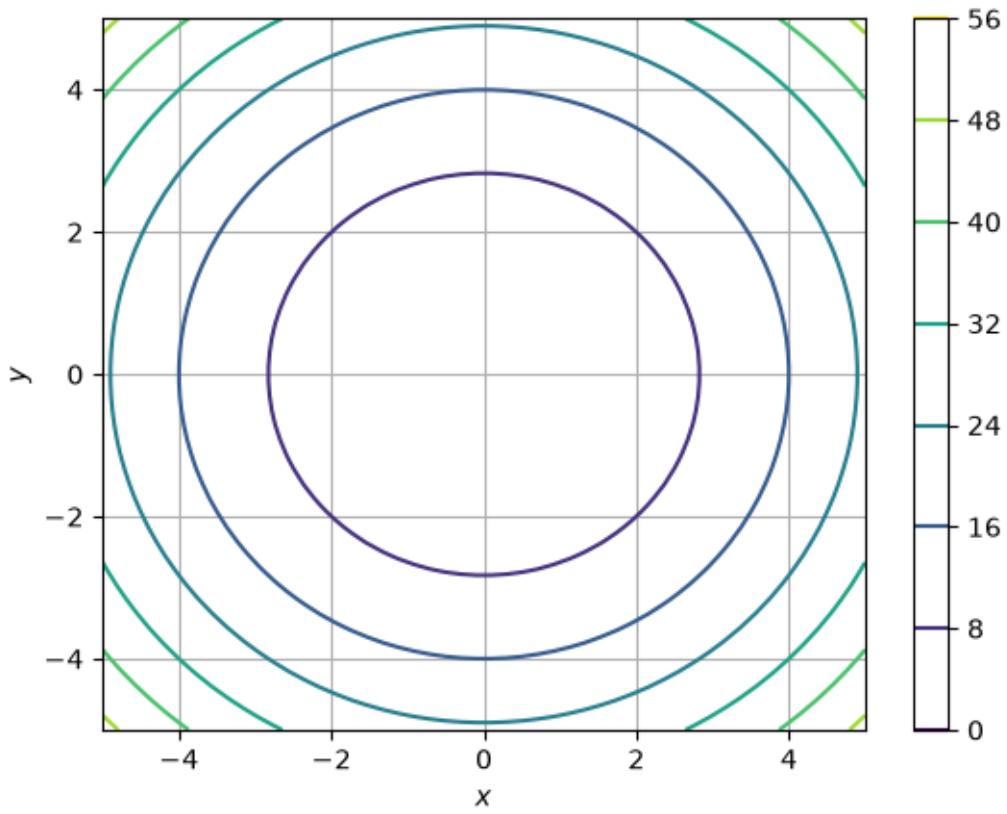
X, Y = np.meshgrid(x, y)

Z = X**2 + Y**2

plt.contour(X, Y, Z)

plt.xlabel(r'$x$')
plt.ylabel(r'$y$')
plt.colorbar()

plt.grid()
plt.show()
```



12. campo vetorial

[46]:
```
x = np.linspace(-5, 5)
y = np.linspace(-5, 5)
X, Y = np.meshgrid(x, y)

R = np.sqrt(X**2 + Y**2)

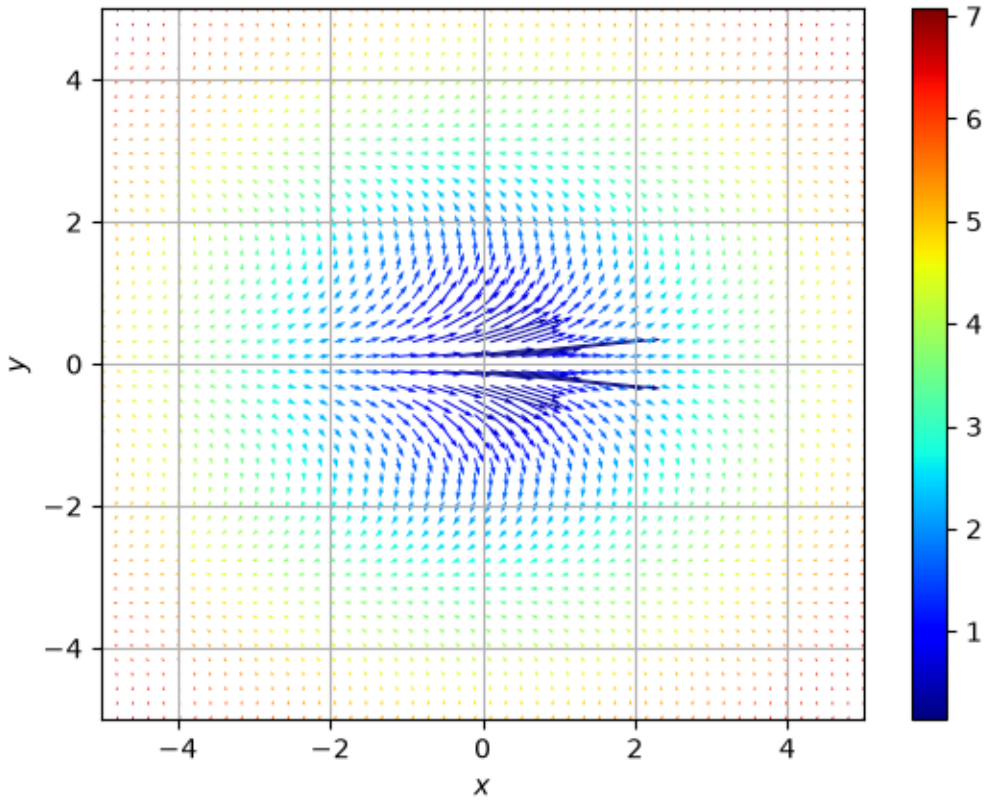
U = np.cos(Y) / R
V = np.sin(Y) / R

fig = plt.figure()

campo = plt.quiver(X, Y, U, V, R, cmap="jet")
fig.colorbar(campo)

plt.xlim(-5, 5)
plt.ylim(-5, 5)
plt.xlabel(r'$x$')
plt.ylabel(r'$y$')
plt.grid()
```

```
plt.show()
```

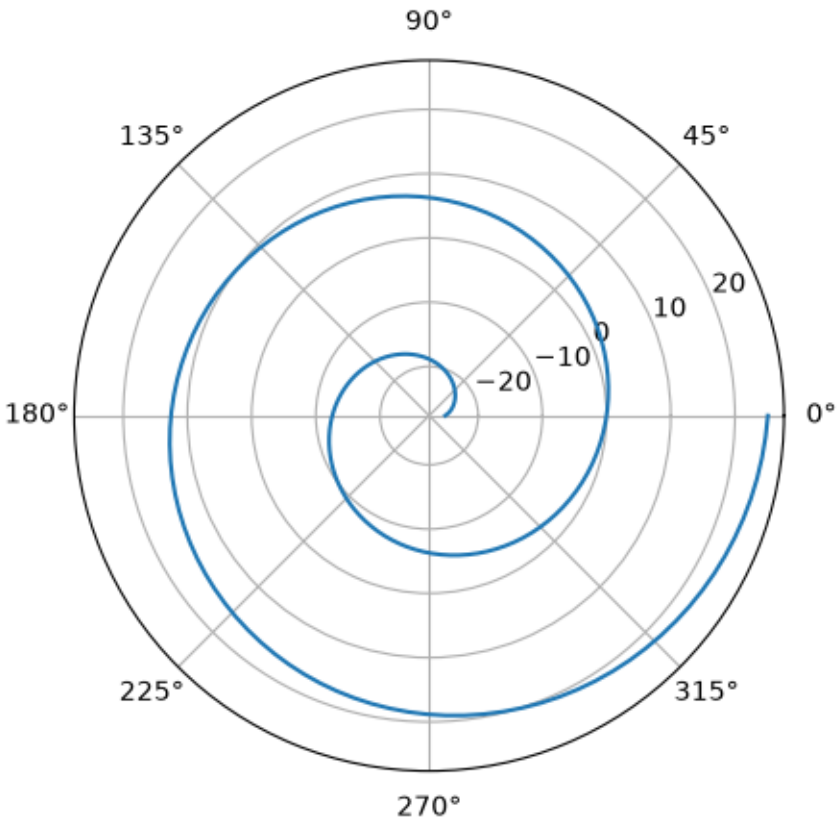


13. gráfico polar

[51]:
```
theta = np.linspace(-2*np.pi, 2*np.pi, 200)
r = 4 * theta

fig, ax = plt.subplots(subplot_kw={'projection': 'polar'})

ax.plot(theta, r)
plt.show()
```



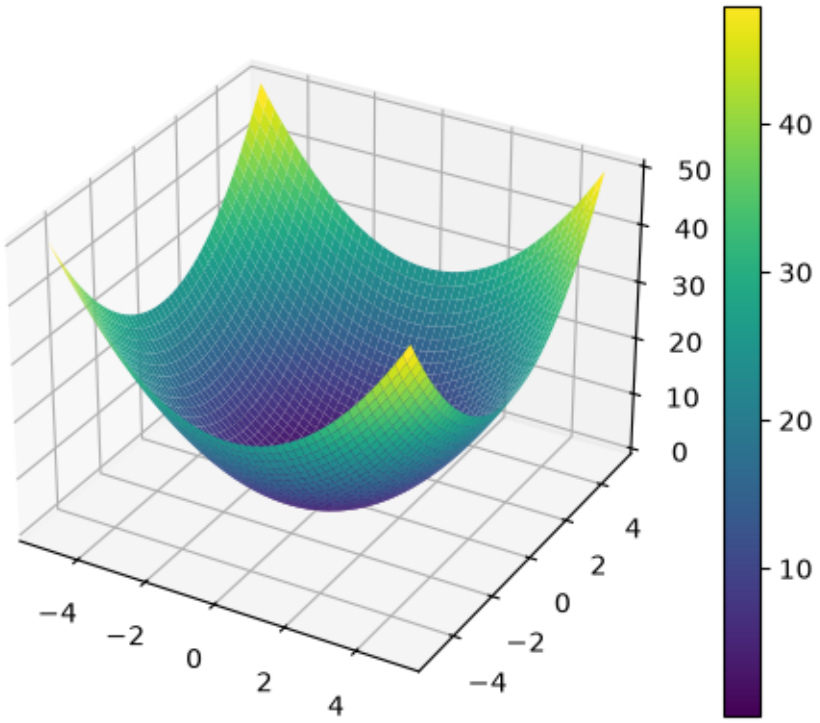
14. superfície 3D

[53]:
```
x = np.linspace(-5, 5)
y = np.linspace(-5, 5)
X, Y = np.meshgrid(x, y)

Z = X**2 + Y**2

fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')

superficie = ax.plot_surface(X, Y, Z, cmap="viridis")
fig.colorbar(superficie)
plt.show()
```



15. comparação de quatro visualizações

[89]:
```
x = np.linspace(0, 2*np.pi)

# cria a figura
fig, ((ax1, ax2), (ax3, ax4)) = plt.subplots(2, 2, figsize=(10, 8))

# primeiro grafico: y=sin(x)+cos(x)
ax1.plot(x, np.sin(x) + np.cos(x))
ax1.set_title(r'$y=\sin(x)+\cos(x)$')

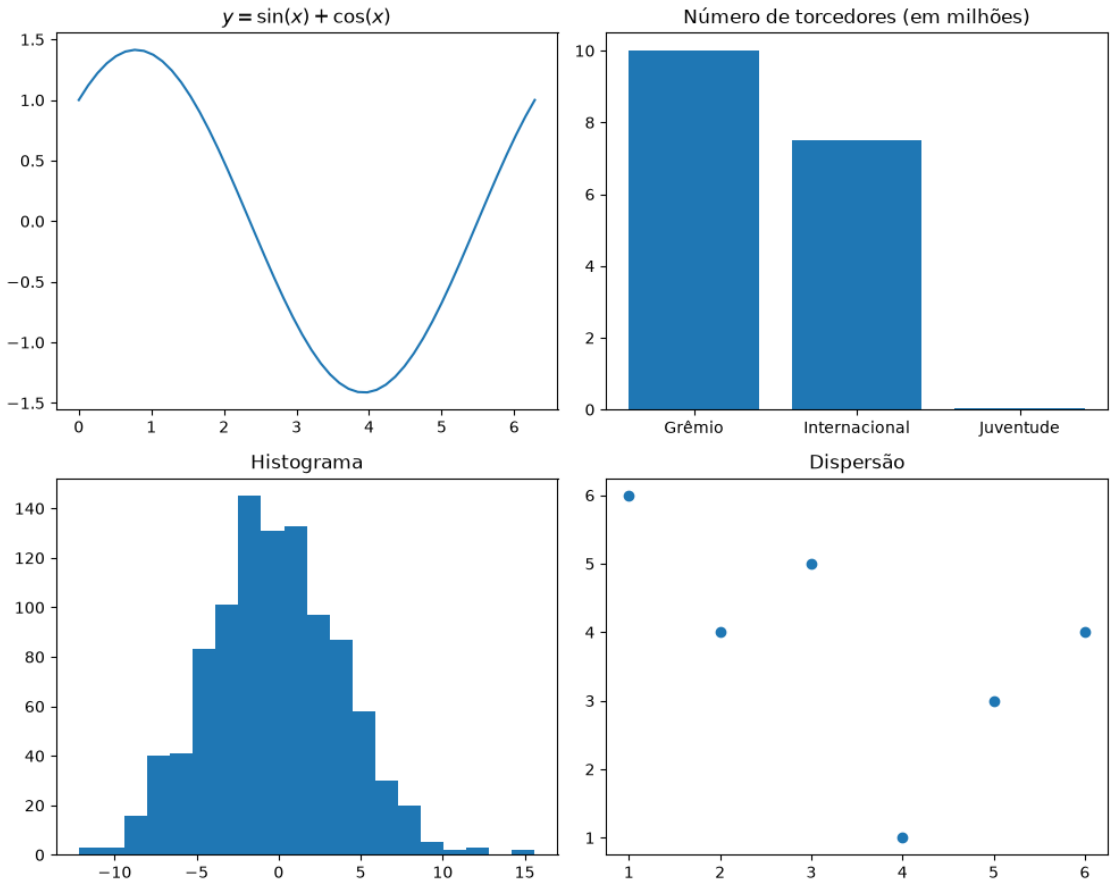
# segundo grafico: barra
times = ["Grêmio", "Internacional", "Juventude"]
torcedores = [10, 7.5, .03]
ax2.bar(times, torcedores)
ax2.set_title("Número de torcedores (em milhões)")

# terceiro grafico: histograma
dados = np.random.normal(0, 4, 1000)
ax3.hist(dados, bins=20)
ax3.set_title('Histograma')

# quarto grafico: dispersao
```

```
ax4.scatter([1, 2, 3, 4, 5, 6], [6, 4, 5, 1, 3, 4])
ax4.set_title('Dispersão')

# plota tudo
plt.tight_layout()
plt.show()
```



16. exercício adicional – gráfico científico completo

[94]:
```
x = np.linspace(-10, 10, 300)
y = np.exp(-0.2*x) * np.sin(x)

plt.figure(figsize=(8, 5))
plt.plot(x, y, label=r'$f(x) = e^{-0.2x}\sin(x)$')
plt.axhline(y=0, color='black', linestyle='--', linewidth=0.8, label=r'$y=0$')

plt.fill_between(x, y, 0, where=(y > 0), color='green', alpha=0.3, label='f␣
↪positiva')
```

```
plt.fill_between(x, y, 0, where=(y < 0), color='red', alpha=0.3, label='f␣
 ↪negativa')

plt.title('gráfico do produto de funções')
plt.xlabel(r'$x$')
plt.ylabel(r'$f(x)$')
plt.grid(True)
plt.legend()

plt.show()
```

